

Process for Field-Wiping a FlashArray

Generated On Fri, 04 Feb 2022 10:24:36 GMT
Confidential - Under NDA

For the latest information, please visit
http://support-sandbox.purestorage.com/FlashArray/FlashArray_Security/FlashArray_Security_Guides_and_Manuals/Remote_Support_Process_for_Field-Wiping_a_FlashArray

© 2015-2022 Pure Storage® ("Pure"), Portworx® and associated its trademarks can be found [here](#) as and its virtual patent marking program can be found [here](#). Third party names may be trademarks of their respective owners. The Pure Storage products and programs described in this documentation are distributed under a license agreement restricting the use, copying, distribution, and decompilation/reverse engineering of the products. No part of this documentation may be reproduced in any form by any means without prior written authorization from Pure Storage, Inc. and its licensors, if any. Pure Storage may make improvements and/or changes in the Pure Storage products and/or the programs described in this documentation at any time without notice. This documentation is provided "as is" and all express or implied conditions, representations and warranties, including any implied warranty of merchantability, fitness for a particular purpose, or non-infringement, are disclaimed, except to the extent that such disclaimers are held to be legally invalid. Pure shall not be liable for incidental or consequential damages in connection with the furnishing, performance, or use of this documentation. The information contained in this documentation is subject to change without notice.



ASP Note

If a FlashArray is to be returned to Pure Storage, as with an RMA or the termination of a customer contract, a field-wipe may be performed at the customer's site in advance of shipping. This ensures that none of the data can be accessed when the device is beyond the customer's control. **To request this service, contact Support.** Only systems that are to be returned to Pure Storage are eligible for this service. It is not available for any other reason, including relocation or re-purposing.

These policies apply to all types of contracts and warranties, including ES2 contracts.



Do not use this process unless absolutely necessary.



This process is used to clean the FlashArray in order to send it back to manufacturing. **Do not transfer a FlashArray from one customer to another using this procedure; all production systems going to customers must first be sent to Pure Storage.**





Internal and Confidential Note



The processes described in this article may be used **only by customers** who are acting under the direction of a Pure Storage Systems Engineer (SE) or an Implementation Engineer (IE).

No ASPs are allowed to use these processes or to direct customers in using them. When performing field wipes, please add to the implementation calendar if it will take an hour or longer.

While performing these processes, **do not** delete customer data. You may only *direct* the customer in deleting data. Contact a Pure Support director or vice president if you believe an exception needs to be made.

In the event that customer data needs to be deleted or altered in whole or just a subset of data due to Pure Storage induced pathologies (SW bug / Corruption / HW Defect etc.), which would require support or engineering to delete customer data, please secure approvals from ONE of the individuals listed below before proceeding. This approval needs to be in place prior to offering customers the option to delete the data.

Flash Array

Boris Zemlyak - VP Engineering FlashArray

Prakash Darji - GM FlashArray

John Colgrove (Coz) - Pure Storage Founder

Problem

Under normal conditions, Pure Storage's policy states that field-wiping of FlashArrays should only be performed by an **onsite** SE or IE, but circumstances may arise where TSEs are required to provide assistance. (This cannot be performed by remote). Then the process may only be performed with the permission *and* the active supervision of a Pure Support manager who will witness and verify the completion of each step as outlined below. When performing field wipes, please add to the implementation calendar if it will take an hour or longer.

SE/IE Checklist

The following is a checklist of all of the tasks to be accomplished in the field-wipe process. The checklist is for your convenience when double-checking your work. The detailed process is presented directly below, under

[Detailed Step-by-Step Process](#).

- The customer provided an official Factory-Reset Request. (The case must be initiated by the customer or the account team that is requesting the erasing of the FlashArray). The FlashArray's name, asset serial numbers, contact, and change-control information are provided in the request.
- Create a separate support case for each FlashArray that requires erasing. Do not duplicate or clone cases.





- The case must be customer-facing.
- The SE/IE confirms the request with the customer and provides guidance on the process, explaining email approvals and other requirements.
- The SE/IE confirms that the asset matches customer request.
- The SE/IE confirms that the customer authorized erasing the FlashArray.
- Each case must be opened a minimum of 24 hours in advance. (Exceptions must be approved by either a director or vice president of the Support Department).
- The customer confirms that all data has been deleted, (preferably 24 hours in advance of erasing the FlashArray).
- The customer confirms via email that no negative impact will occur as a result of the erasing process.
- If necessary, the SE/IE connects to the FlashArray using VATS.
- The SE/IE confirms that the FlashArray is *not* in use and that no sessions with any other device are active; there is no traffic of any kind on the FlashArray.

The next steps are accomplished under “three sets of eyes,” meaning the SE/IE requires two witnesses for all of the following actions. The witnesses can include a Pure Forensics team member, a Pure Support Manager, an on-site ASP technician, an on-site technician, or an on-site manager of technicians.

- The SE/IE and witnesses confirm that they are working on the right array, verifying the FlashArray name, serial numbers case number, account name and customer's written approval documented in Salesforce.
- The SE/IE and witnesses verify that the FlashArray that they are working on has no customer data.
- The SE/IE and witnesses update the Salesforce with approval to proceed.
- The customer generates a token for the Support Department to reset the FlashArray, (if applicable).
- The SE/IE logs in as root and then directs the customer in issuing commands, opening a Zoom session, if applicable.
- The SE/IE and witnesses update the customer on the status of the factory reset. Verbally in the Zoom session *and* through email confirm that no unintended impact has resulted because of the wipe.

TSE Checklist

The following is a checklist of all of the tasks to be accomplished in the field-wipe process. This checklist is for your convenience when double-checking your work. The detailed process is presented directly below, under [Detailed Step-by-Step Process](#).

- The customer provided an official Factory-Reset Request. (The case must be initiated by the customer or the account team that is requesting the erasing of the FlashArray). The FlashArray's name, asset serial numbers, contact, and change-control information are provided in the request.
 - Create a separate support case for each FlashArray that requires erasing. Do not duplicate or clone cases.
 - The case must be customer-facing.
 - The TSE confirms the request with the customer and provides guidance on the process, explaining email approvals and other requirements.
-





- The TSE confirms that the asset matches customer request.
- The TSE confirms that the customer authorized erasing the FlashArray.
- Each case must be opened a minimum of 24 hours in advance. (Exceptions must be approved by either a director or vice president of the Support Department).
- The TSE confirms via email that the customer authorized erasing the FlashArray.
- Each case must be opened a minimum of 24 hours in advance. (Exceptions must be approved by either a director or vice president of the Support Department).
- The customer confirms that all data has been deleted, (preferably 24 hours in advance of erasing the FlashArray).
- The customer confirms via email that no negative impact will occur as a result of the erasing process.
- If necessary, the TSE connects to the FlashArray using VATS.
- The TSE confirms that the FlashArray is *not* in use and that no sessions with any other device are active; there is no traffic of any kind on the FlashArray.

The next steps are accomplished under “four eyes,” meaning both the TSE and a Support Department manager who is present for all actions.

- The TSE and manager confirm that they are working on the right array, verifying the FlashArray name, serial numbers case number, account name and customer's written approval documented in Salesforce.
- The TSE and manager verify that the FlashArray that they are working on has no customer data.
- The TSE and manager update the Salesforce with approval to proceed.
- The customer generates a token for the Support Department to reset the FlashArray, (if applicable).
- The TSE logs in as root and opens a Zoom session and then directs the customer in issuing commands.
- The TSE and manager update the customer on status of the factory reset. Verbally in the Zoom session *and* through email confirm that no unintended impact has resulted because of the wipe.

Detailed Step-by-Step Process

The following details a step-by-step process for field-wiping a FlashArray. For convenient reference, technical details on sub-processes are presented below the detailed steps.

Preliminary

If the customer has not yet logged a request for field-wiping the FlashArray, inform them of the requirement. They must provide explicit, written approval from an authorized contact that documents the following information:

- FlashArray name, including domain, (eg. US1-Pure7942.cscinfo.com)
- Asset serial numbers for controllers, chassis and shelves.
- Approval for erasing the FlashArray.





- Date and time of the change-control window for the erasing process.
- A customer contact for the duration of the change-control window.

Share the template and the instructions for filling out the request with them. (See [Filling in a Factory-Reset Request Template](#), below).

1. Create a separate support case for each FlashArray that requires erasing. Each support case must be opened for a minimum of 24 hours in advance. If 24 hours have not passed from the opening of the support case, get approval for the exception from either a director or a vice president of the Support Department.
 - Do not duplicate or clone cases.
 - The case must be customer-facing.
 - Each case must be initiated by either the customer or the account team.
2. Over phone or email, confirm with the customer the request, the date, and the time of the change-control window. Then provide guidance for them on how the process will proceed. Make sure to inform the customer that, prior to the change control window, preferably 24 hours in advance, the following needs to occur or the process cannot go forward. Provide specific instructions as necessary. (See [Checking for Replication](#), below).
 - All volumes, snapshots, and hosts need to be deleted.
 - Replication needs to be disabled.
3. Confirm that the asset matches the account name and the asset serial numbers on the official request for the field wipe. To do so, use FUSE to verify the serial numbers by entering `purehw list --all`. Alternately, use Salesforce to confirm serial numbers by clicking on the account name then searching for them under the section titled *Assets*.
4. Confirm that the customer contact is authorized to approve erasing the FlashArray.
5. Ask the customer to confirm that no negative impact to the environment will occur as a result of the erasing process.
6. Once all of the requirements listed above have been met and you are operating within the date and time of the change-control window, connect to the FlashArray using VATS. Pay strict attention to each of the following restrictions (If you are onsite and able to use a local console, skip to Step 7):
 - The customer must be present on a Zoom live video session, (which should be recorded if the customer agrees with your request).
 - Only use VATS to connect via phone or email.
 - Close all other Remote Assist and VATS sessions.
7. Confirm that all volumes, snapshots, and hosts have been deleted and that replication has been disabled, entering the following commands. If any volumes, snapshots, or hosts remain, or if replication has not been disabled, return to Step 2.
 - For volumes on versions through 4.10+, enter `purevol list`; for versions beginning 5.0+, enter `purevol list --pending`.
 - For Snapshots on all versions, enter `purevol list--snap`.
 - For hosts on all versions, enter `purehost list`.





8. Before any commands are executed, have your witness/manager confirm again with you, that you are working on the correct device and have explicit approval by checking the FlashArray name, serial numbers, case number, account name, and the customer's written approval as documented in the case.
9. With your Manager update the support case.
10. If the system is Purity//FA 4.8.x+, ask the customer to generate a token that can be used to reset the FlashArray as follows:
 1. Ask the customer to run this command from the Pure user level: `purearray authorize factory-reset --eradicate-all-data`.

```
pureuser@vm-quelyn5> purearray authorize factory-reset --eradicate-all-data
Name      Authorization Code
factory reset 4671919
```



In Purity 5.2+, currently, the above command will not work, displaying the following message.

```
pureuser@slc-m70> purearray authorize factory-reset --eradicate-all-data
Error: Missing or invalid parameter.
```

This is due to PURE-147293. If you are on 5.2, skip to step 10 b and turn off Purity on both controllers via `pureadm stop` (stop after each controller, making sure you are on the correct array.)

If you are in a situation where Purity//FA is not running and it can't be started, use the following command instead:

```
apartment_reset_authorization_helper --authorize ARRAY_NAME --delay 0
```

(The above command must be run as root and not as pureuser).

2. With either command, the system generates an authorization token that can be used immediately, (delay 0). The authorization will take effect 10 minutes after the command is run. It will then remain in effect for 24 hours.
 3. Ask the customer to log out of the session where they generated the authorization code and then log into a new session. Otherwise, the code will not work. Note that a short delay is advised (approximately ten minutes) before logging into a new session.
11. Assist the customer in logging into root. For questions, see the [Dynamic Root Password](#) procedure.





At this point, **do not** execute any commands on the FlashArray. Allow the customer to make all entries. *Only* the customer can be responsible for actions on the device.

12. Instruct the customer to execute commands in one of the two following sections as you and the other witness watch:
[Resetting the FlashArray \(4.8.0+\)](#)
[Resetting the FlashArray \(4.7.10 and Earlier\)](#)
At this point, *only* the customer should execute any of the remaining commands on the FlashArray.
13. Update the case with appropriate notes and with screenshots of command-line output, including the timestamp. (see [Gathering Information for Certification of Erasure](#)).
14. Instruct the customer on the physical shutdown, removal, and the return of the FlashArray to Pure Storage, (see [Shutting the System Down for Removal](#)).
15. Instruct the customer and provide documentation on obtaining certification for the field wipe and erasure of the FlashArray, (see [Certifying the Completed Wipe of the FlashArray](#)).
16. Over a live Zoom session with your witnesses and the customer, update the customer on the completion of the field wipe process of the FlashArray. Confirm that no negative impact has occurred within the environment. Read through the requirements of certifying the field wipe, (see [Certifying the Completed Wipe of the FlashArray](#)).

Filling in a Factory-Reset Request Template

The following template is to be submitted in a Salesforce case. To fill in the Factory Reset Request Template, replace the series of Xs at each occurrence with the correct information.

Finding Serial Numbers for the Assets

The serial number used for the identification of the asset may be problematic because different models present different identification options and occasionally some components, such as controllers or chassis, may have been replaced during the life of the FlashArray. To find serial numbers to use in the Factory-Reset Request and which the technicians can use to verify they are working on the correct device, see the following customer instructions.

Customer Instructions

1. Use SSH log into either controller's management IP address or vir0 IP address.
2. Log in as pureuser.
3. Run `purehw list --all`. Text similar to the following displays.

```
pureuser@SLC-m20> purehw list --all
Name  Status  Identify  Slot  Index  Model [...] Type      Handle                                Parent
CH0   ok      off       -     0      [...] chassis    M_SERIES_PCHFS15280023  hwroot
CT0   ok      off       -     0      [...] controller  PLATSASA_PCTFS152500BC  M_SERIES_
```





```
PCHF515280023
CT1  ok   off   -   1   -   [...] controller  PLATSASA_*PCTFS152500B0*   M_SERIES_
PCHF515280023
SH0  ok   off   -   0   -   [...] storage_shelf EB-2425-E12EBD_*SHG1019220G57BA* hwroot
```

The same serial numbers as those listed can be found on the back of the chassis, controllers, and shelves.

Factory Reset Request Template

Use the below customer facing language.

Dear xxxxxxxx,

Thank you for reaching out to Pure Storage Support. This is to acknowledge your request, Pure Storage Support Case xxxxxxxx to factory reset a Pure Storage FlashArray/FlashBlade. Listed below are the details for the Array, which requires a factory reset, and the timings for the change control window. Kindly, confirm via email the information listed below is accurate and your consent to proceed with the scheduled change control.

Array name including domain (FlashArray) and/Array ID (FlashBlade) - XXXXXXXX

Array ID or Controller Serial numbers - XXXXXXXX

Date/Time of the change control for the wipe to take place - XXXXXXXX

Customer contact for the duration of the change control - XXXXXXXX

Additionally, Pure Storage requires that all customer data (volumes and hosts attached to the array, replication disabled, snapshots removed) be deleted prior to the change control window by [insert customer company name].

We strongly recommend that this deletion takes place 24hrs in advance of the scheduled activity. Pure Storage also recommends eradication of data to take place during the change control window if it's not automatically eradicated after 24hrs of being deleted. Pure Storage will not be able to proceed with the scheduled activity if any customer data is present on the array.

Lastly, please reply to this email with your consent to proceed with the factory reset of the Pure Storage FlashArray/FlashBlade [insert array name] on [Month, Date, Time]. Pure Storage will send out a Zoom invitation for you to join during this change control.

Thank you

Pure Storage Support





Checking for Replication

If you have replication configured on the FlashArray to which you intend to field-wipe, you need to remove it before proceeding. If you do not, there will be orphaned references on the previously paired FlashArray, which can lead to orphaned pgroups.

1. Check if the FlashArray is connected to any peer by entering the `purearray list --connect` command. (This information will also display in `diagnostics.log`).

```
$ purearray list --connect
Name          ID                      Version Management Address Replication Address Connected
Type
mv-sup-fa405  e50bd631-b29b-82b1-d0e6-5bf929a5b223 -    10.14.84.35      10.124.223.231
True          replication
```

2. If the connected status returns as `True`, run the following command to disconnect from the peer FlashArray:

```
purearray disconnect --local-only <peer_array>
```

3. Check to make sure there are no snapshots from the peer FlashArray by running the following command. If a listing of snapshots is returned, instruct the customer to remove them. (If they want to keep them, they need to be copied to a different volume).

```
purepgroup list --snap | grep "<peer_array>:"
```

4. Check if the peer FlashArray is listed with the following command:

```
purearray list --connect
```

5. To "forget the peer," run the following command:

```
puredb replication forgetpeer <peer_array>
```

Resetting the FlashArray

Resetting the FlashArray (4.8.0+)

As of Purity version 4.8.0, resetting FlashArrays requires running the `reset_apartment` command.

As a prerequisite, the procedure requires the customer to create a token on the FlashArray in order to authorize Pure Storage to proceed with the resetting process, (see Step 11, above, under [Detailed Step-by-Step Process](#)). The `purearray authorize` command should be run only by the customer. If for some reason they cannot or will not run the `purearray authorize` command, contact Support Escalations.

Additionally, this procedure requires Purity software to be running. If Purity is *not* running, contact Support Escalations.





To reset the FlashArray, do the following:

1. Ask the customer to run the `hostname` command in order to provide the name of the system on which this is being run. This is necessary for the certificate below:

```
root@mv-sup-fa420-ct1:~# hostname
mv-sup-fa420-ct1
```

2. Stop all activity and wait 10 minutes before proceeding. To do so, ask the customer to enter `pureadm stop` on both controllers. The authorization will take effect 10 minutes after the command is run. It will then remain in effect for 24 hours.
3. Ask the customer to run `reset_apartment -A -T <token>`:

```
root@vm-quelyn5-ct0:~# reset_apartment -A -T 4671919
Successfully reset 24 files.
Apartment successfully created.
Sep 29 06:17:32.054 begin malloc stats
Arena 0:
system bytes   =    135168
in use bytes   =         0
Arena 1:
system bytes   =   1073152
in use bytes   =   1071504
Arena 2:
system bytes   =  48267264
in use bytes   =  48263824
Total (incl. mmap):
system bytes   =  49475584
in use bytes   =  49335328
max mmap regions =         0
max mmap bytes  =         0
Resetting files
```

4. For further instructions, return to Step 12 above, (under [Detailed Step-by-Step Process](#)).

Resetting the FlashArray (4.7.10 and Earlier)

As a prerequisite, volumes should already be deleted, as described in Step 7, (under [Detailed Step-by-Step Process](#)). If they are not, make sure that the customer does so at this time. Additionally, the procedure requires the customer to create a token on the FlashArray in order to authorize Pure Storage to proceed, (see Step 10, above). After the token is generated, the customer must log into a new session before you give them root access; (see Step 11, above).

To reset the FlashArray, do the following:

1. Ask the customer to run the `hostname` command in order to provide the name of the system on which this is being run. This is necessary for the certificate below:
-





```
root@mv-sup-fa420-ct1:~# hostname  
mv-sup-fa420-ct1
```

2. Ask the customer to run `pureadm stop` on *both* controllers before proceeding.
3. Ensure that services are offline using `pureadm status` on both controllers. All processes should say "stop/waiting" or "not responding":

```
# pureadm status  
Process status:  
purity stop/waiting  
lio-driv stop/waiting  
cached stop/waiting  
foed stop/waiting  
platform stop/waiting  
gui stop/waiting  
rest stop/waiting  
monitor stop/waiting  
iostat stop/waiting  
statistics stop/waiting  
platform not responding  
foed not responding  
gui not responding  
platform_env not responding  
foed_env not responding  
remote_patch not responding  
driver not responding  
san not responding  
health not responding
```

4. If the customer is on version 4.5.17 or below after Purity is stopped on CT0, drop and recreate the database. This step overwrites the database. To do so, enter `drop_db` and `create_db` commands, as follows. (If you are running 4.6.x+, skip this step, and proceed to step 5).

```
/opt/Purity/bin/db.py drop_db  
/opt/Purity/bin/db.py create_db
```

5. Ask the customer to run the `ssd_reset -A` command to start the reset process:

```
ssd_reset -A -e -t
```

Important: Running the reset with `-A`, as shown above, is required so that the tool does not create a new apartment. If the shelf is added to a system at a later time, `puresetup` will prompt creation of the apartment, if needed.

Once completed, the system notifies you that it successfully "Reset X Devices." The notice changes, depending on how many SSDs are in the system.

```
root@vm-quelyn-ct0:~# ssd_reset -A -e -t
```





Resetting devices
Successfully reset 24 devices.

6. For further instructions, return to Step 12 above, (under [Detailed Step-by-Step Process](#)).

Gathering Information for Certification of Erasure

To gather information for the certification of the erasure, do **one of the following** depending on the type of Erasure:

1. If this is a **full array erasure**, take a screenshot of your session that includes the following:
 - Hostname
 - `ssd_reset` or `reset_apartment -A -T`
 - Reset devices
2. Or If the certificate is needed only for a **shelf or datapack evacuation** (not a full array wipe), we can collect evidence of erasure with the following log lines once the data on the drives is fully evacuated:
 - For 4.7.x, collect the output of the below including full path & syntax:
`zgrep "ha::secure_erase.success" platform.log*`
 - 4.8.x and above, collect the output of the below including full path & syntax:
`zgrep -i "Device secure erased" platform.log*`

Certifying the Completed Field Wipe of the FlashArray

If the customer requests certification of the array wipe, the technician who performed the wipe should send the following information in a request to eaops@purestorage.com. Below are the required items to get an Erasure Certificate:

- Customer Name.
- FlashArray Name(s).
- Screenshots of `ssd_reset` or `reset_apartment -A -T` output (for array return) or the log lines captured from `platform.log` (for capacity consolidation).
- Full Name of the person who executed the erasing procedure (customer employee name).
- The date that the erasure procedure was executed.



Once the Field Wipe of the FlashArray has been completed and verified (by examining the logs), perform the following procedure:





1. Push phonehome logs:
(See [Upgrade Guide - Hardware NDUs](#)).
 1. Push the phonehome logs, for both controllers: `purearray phonehome --send-today && ssh peer purearray phonehome --send-today`
 2. Check the status of push: `purearray phonehome --status && ssh peer purearray phonehome --status`
 3. Logs successfully pushed will give this output:

```
root@some-array-ct1:~# purearray phonehome --status && ssh peer purearray
phonehome --status
Status Action
- - - -
Status Action
- - - -
```
2. Then the logs themselves should be wiped by executing "puresetup cleanup". This should be completed before initiating the final step "Shutting the System Down for Removal".

Shutting the System Down for Removal

To shut down after you have proof of the erasure for certification, pull the power cords from the system, and instruct the customer that the system needs to be un-racked. Provide documentation as necessary.

